

Asbah Talal

Website (in progress): asbahtalal.github.io/blog | asbah.talal@gmail.com | (405) 433-7446

Skills: SolidWorks, Fusion 360, MATLAB, Python, JavaScript, Java, C++, SQL, HTML/CSS, UI/UX, Arduino, Raspberry Pi, React, FastAPI, OpenCV, NumPy, Pandas, Git/GitHub, VSCode Linux, Embedded Systems, Sensor Integration, CAD, 3D Printing, Soldering, Excel

EDUCATION

Texas A&M University - College Station, TX

Dec 2028

- *BS Mechatronics Engineering, pursuing minors in Computer Science and Arabic* | GPA: 3.7
- *Fencing Athlete*

RELEVANT EXPERIENCE

Texas A&M Turtle Robotics Advanced Research Projects

Aug 2025 - May 2026

Disaster Response Observation Network (DRON) Mechanical and Software Project Member

College Station, TX

- Developed an autonomous disaster-response drone system integrating thermal and LiDAR sensors to map rapidly changing environments
- Transmitted real-time, high-accuracy data to first responders to support faster and safer decision-making
- Redesigned sensor mounts and mechanical components with Solidworks to improve stability, accuracy, and reliability during flight

Optical Rotation and Imaging Observation (ORIO) Electrical and Software Project Member

College Station, TX

- Designed and developed a 3D-printed equatorial star-tracking mount for astrophotography applications using stepper motors and microcontroller-based control systems
- Modeled assemblies and systems in SolidWorks, including cycloidal drive and belt-driven rotation mechanisms
- Integrated electrical components including motor drivers, power regulation systems, and ESP32/Arduino-based control hardware for automated sidereal-rate tracking
- Collaborated on software and hardware systems enabling autonomous tracking, calibration, and future GoTo positioning functionality for celestial object navigation

Autonomous Underwater Vehicle (AUV)

Aug 2025 - May 2026

Programming & Mechanical Team Member

College Station, TX

- Developed control systems and mechanical improvements for Texas A&M's autonomous underwater robot
- Implemented machine-vision pipelines and Doppler Velocity Logger navigation systems
- Designed actuation behaviors for RoboSub challenges including gate traversal and octagon surfacing

NASA

June 2024 - Aug 2024

NASA CSR (Center for Space Research) SEES Air Quality Intern

Virtual

- Analyzed nationwide air-quality conditions using PurpleAir sensor networks and NASA atmospheric datasets, applying Python-based statistical modeling to identify pollutant trends across urban and rural regions.
- Developed predictive air-quality models by processing high-volume environmental data (PM2.5, ozone, humidity, temperature) to evaluate pollution behavior and assess potential health and climate impacts.
- Selected to present research findings at the 2024 American Geophysical Union (AGU) Conference

NASA Cubes in Space

Aug 2022 - July 2023

Science Museum Oklahoma

Oklahoma City, OK

- Designed a radio-frequency (RF) transceiver payload selected to fly on a [NASA scientific balloon](#).
- Created detailed engineering drawings and component layouts to meet strict mass, volume, and thermal constraints.
- Integrated RF components and sensors into a structure optimized for atmospheric pressure and temperature changes.
- Analyzed post-flight RF data to assess how radio waves behaved at high altitude
- Collaborated with NASA mentors and program engineers to refine the payload design and verify flight readiness

StellarXplorers

Oct 2024 - Feb 2025

Competitor

Edmond, OK

- Qualified for the national semifinalist round (top 30%) two years in a row for a satellite mission design competition.
- Determined optimal satellite orbits using space operations tools to meet mission constraints and scenario objectives.
- Analyzed datasets in Excel to select optimal satellite components based on mass, power, and mission constraints.

Technology Student Association

Apr 2023 - Apr 2025

State Region Representative, State Treasurer

Stillwater, OK

- Led and represented 31,000+ Oklahoma TSA members at local, state, and national conferences and events.
- Presented student initiatives at the [Oklahoma State Board of Career and Technology Education Meeting](#).
- Served on the Website, Social Media, Fundraising, and Video Committees, supporting state-level outreach and operations.
- Hosted leadership workshops for chapters statewide to strengthen officer development and competitive readiness.

- Raised funds for the American Cancer Society by coordinating statewide chapter fundraising efforts.

Awards/Projects

Hacklahoma Winner:

2026

[PetPulse](#) | (React, Python FastAPI, PostgreSQL)

- Built an AI-powered pet health monitoring platform as part of our submission to a hackathon with 225 participants
- Integrated sensor, audio, and image analysis pipelines to generate real-time behavioral insights
- Implemented cloud-based event notifications, vet history management, and media analysis tools

TAMU Ideathon Winner:

2025

[Kinect](#) | (Figma, User Research, Wireframing, User Flows, Prototyping, Time Management, Teamwork and Collaboration)

- Kinect is an educational platform for children aged 5-12 that encourages hands-on learning and creativity across topics such as art, writing, life skills, and science. It connects children with a global community in a safe, AI-free environment, allowing them to share their work, collaborate, and learn through doing rather than consuming.
- This project prototype was developed in under 24 hours as part of a hackathon-styled competition called Ideathon.

TAMU HowdyHack:

2025

[Campus Chaos](#) | (Python, Time Management, Teamwork and Collaboration)

- As freshmen, we each experienced a significant learning curve trying to navigate campus. In a fun and engaging way, we wanted freshmen to become familiarized with the different landmarks throughout campus. We were heavily inspired by the Google Dinosaur game to create a fun solution to get familiar with some buildings. Reveille, our mascot, is a major symbol of pride at Texas A&M, so we decided to have her as our main character.

Data Science & Analytics 1st Place:

2023, 2025

[Impacts of Urban Infrastructure](#) | (Research, R, Statistics, Critical thinking, Visualization)

- Placed 1st twice in a state-level data science competition as well as placing 7th at the national level, the highest level of competition in the Technology Student Association, an organization that includes over 300,000 students.